

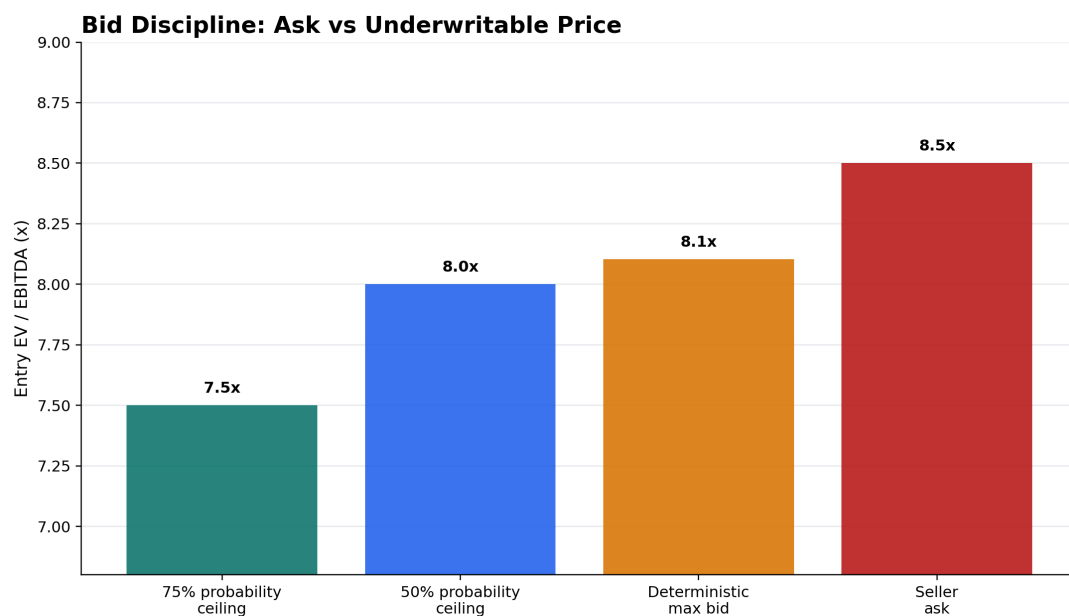
Take-Private Reverse LBO & Probabilistic Underwriting

Project Redwood — investment committee overview

A private-equity underwriting case study designed to answer one practical question: what is the highest price a sponsor can pay while preserving return hurdles, debt capacity and downside protection?

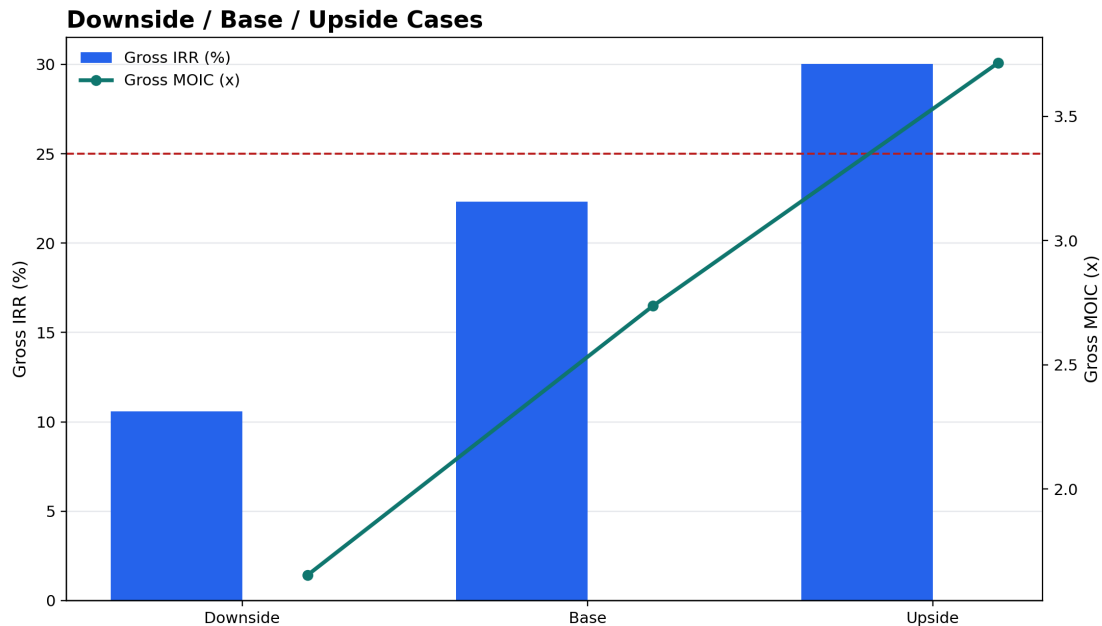
Seller ask	8.5x EBITDA
Base-case gross IRR / MOIC	22.3% / 2.74x
Deterministic max bid	8.1x
P(IRR ≥ 25%) at ask	32.9%
50% / 75% probability bid ceilings	8.0x / 7.5x

The seller asks 8.5x EBITDA, but the deterministic case only clears 22.3% gross IRR. The underwriting therefore becomes a bid-discipline exercise rather than a model that simply forces the deal to work.



1. Base case and bid discipline

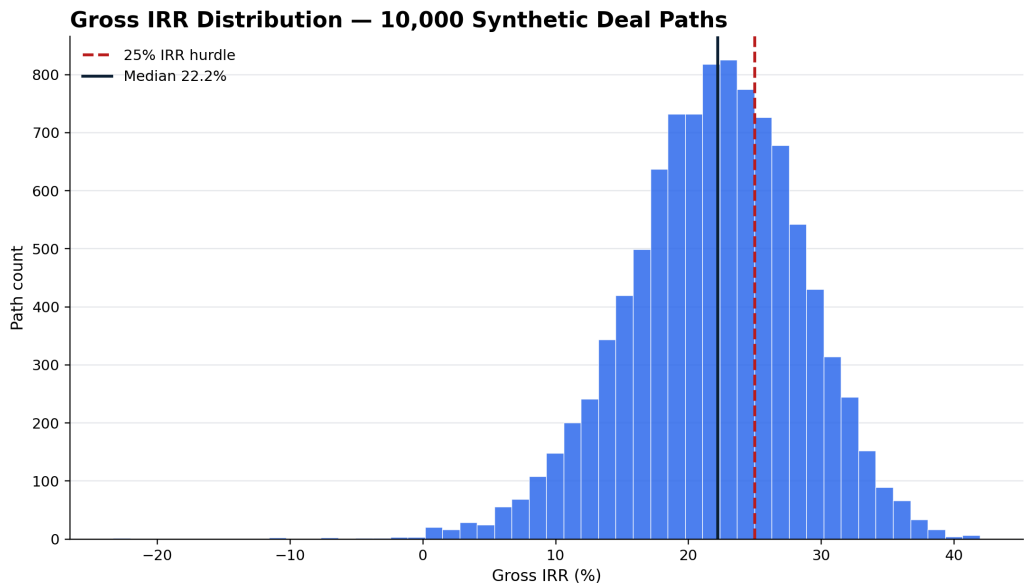
The deterministic underwrite assumes a five-year hold, debt-funded acquisition and gradual EBITDA margin expansion. The reverse LBO converts the target return hurdles into a maximum sponsor equity cheque and, from there, a maximum underwritable entry multiple.



At the seller ask of 8.5x, the model generates 22.3% gross IRR and 2.74x MOIC. The reverse LBO implies a deterministic maximum bid of 8.1x, leaving a bid gap of -0.4x.

2. Probabilistic underwriting

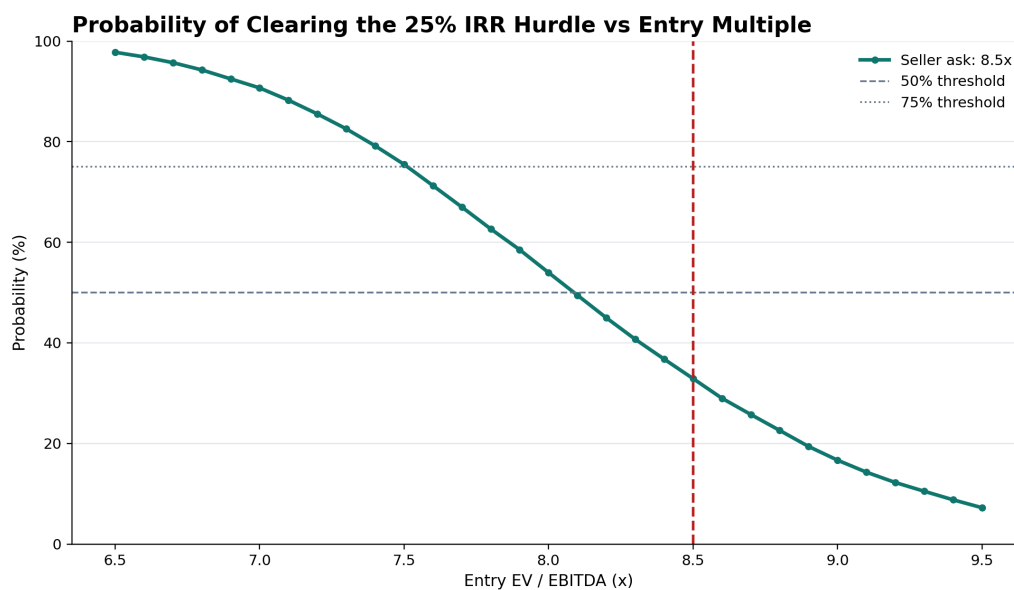
A synthetic Monte Carlo engine varies revenue growth, margins, rates and exit multiples in a correlated way, then runs each path through the same debt waterfall. This changes the discussion from a single point estimate to a probability distribution of outcomes.



At the seller ask, only 32.9% of simulated outcomes clear the 25% gross IRR hurdle. The dashboard therefore also reports risk-adjusted bid ceilings based on target probabilities of success.

3. Probability-based bid ceilings

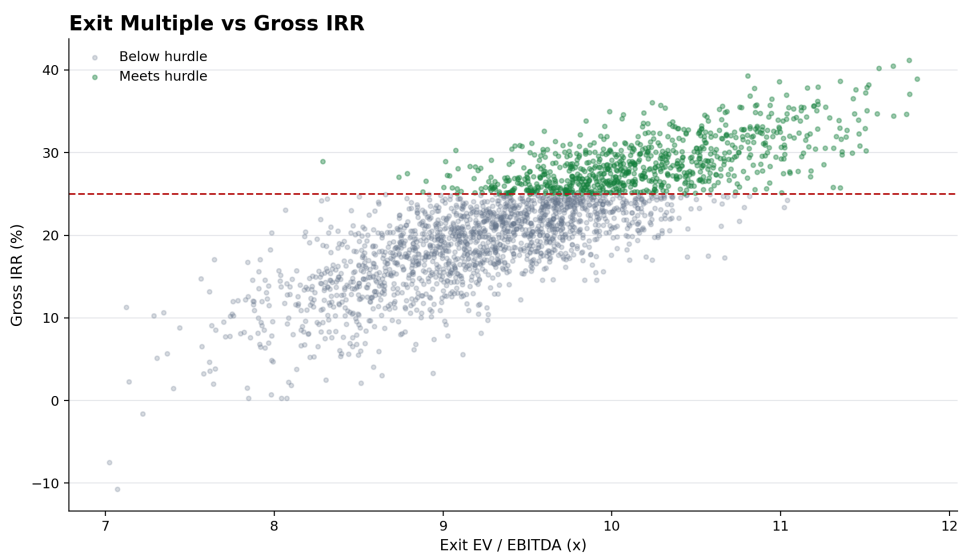
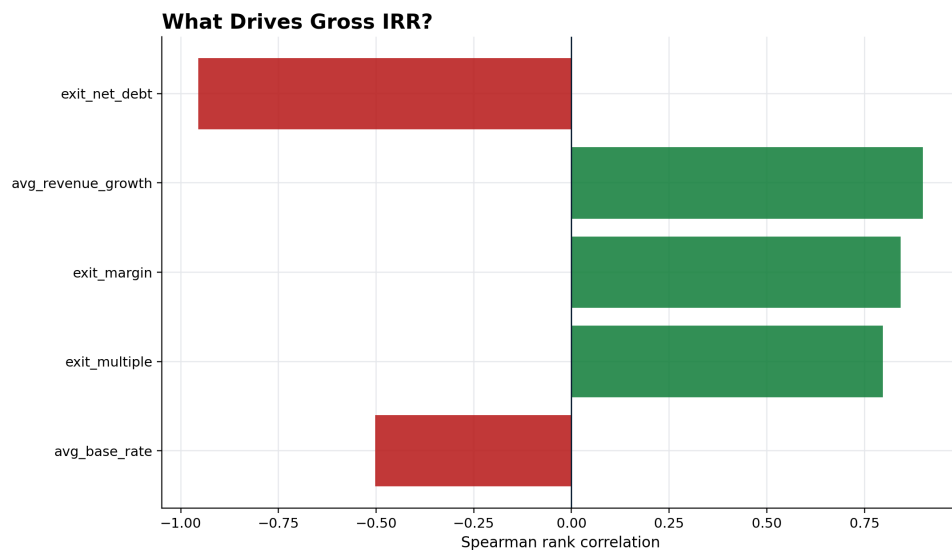
The bid-probability curve converts the simulation results into an underwriting rule: for every entry multiple, what is the probability that the deal clears the target IRR?



The curve suggests a 50% probability bid ceiling of approximately 8.0x and a 75% probability bid ceiling of approximately 7.5x.

4. What drives returns?

The rank-correlation view helps explain the model in interview terms. Higher exit multiples, stronger revenue growth and wider exit margins support returns, while higher average base rates and higher exit net debt drag the distribution down.



5. Deliverables and usage

The repository contains an Excel LBO model, a Python underwriting engine, automated tests, this PDF overview and a browser dashboard where you can change assumptions and regenerate the decision graphics directly in the client.

Main files

- Take_Private_Reverse_LBO_Probabilistic_Underwriting_Model.xlsx
- outputs/interactive_ic_dashboard.html
- run_risk_simulation.py
- tests/test_risk_engine.py

Interview takeaway

This is not just an LBO template. It demonstrates transaction mechanics, bid discipline, downside framing and the ability to move from a single deterministic case to probability-based underwriting.

All data and outputs are synthetic and produced for educational use only.